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Maths puzzles for early finishers

Early Finisher Puzzles

Primary 5

Semester 1

Terms 1 and 2

20 single-page puzzles
Almost every puzzle checks itself
One star and two star pages
Teacher answer key included

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Notes

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The mathematics syllabus is published by the Ministry of Education, Singapore.

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Teacher notes (DRAFT)

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What this is

Twenty single-page maths puzzles for Primary 5 pupils who finish their work early. This is Semester 1 of two booklets (Terms 1 and 2). Each page stands alone, so you can hand out one page or keep a tray of pages. The pupil needs no help to start.

Which terms

The pages are grouped by the order in which Primary 5 topics are typically taught: this booklet follows Terms 1 and 2, the other booklet follows Semester 2 (Terms 3 and 4). Schools vary, so check your scheme of work: use a page whenever its topic has been taught, from either booklet.

Stars

One star pages practise a topic as it is usually taught (10 pages). Two star pages use the same topics with two steps or two clues to combine (10 pages).

How the pages check themselves

The boxed strip at the foot of each page tells the pupil how to check. The check shows that something is wrong. It does not give the answer, and it asks the pupil to see whether the answer makes sense. Every answer box and every letter box must be filled: a box left blank gives no letter, and the check cannot see it.

- **Code word.** The answers give letters that spell a word. The strip gives no hint of the word. Each word can be spelt from its letters in one way only, so answers in the wrong boxes never make another word, and one or two wrong answers very rarely do. A box left empty gives no letter and is not seen by the check.
- **Colour by code.** Each answer is found in a key of five numbers and the part is coloured with that crayon. The pupil counts the parts of each colour and compares with the box on the page; matching parts must share a colour.
- **Route.** The line must reach FINISH through the stated number of boxes, and every box on it must follow the rule on the page.
- **Letters in an order, a grid or a choice.** Each number, card, picture or choice carries a letter; the letters, read in the order asked, must spell a word. Where one item is left out, its letter is a decoy that is not part of the word.
- **Total.** Every row, column and diagonal of the magic square must make the total printed on the page.
- **Loop.** A chain of circles joined by arrows goes all the way round; the last arrow must bring the pupil back to the start number.
- **Both sides equal.** The pupil puts each number back into its box and checks that both sides of every line are equal. A blank box makes the line untrue.

Marking is not needed. To spot-check, look only at the check strip.

Teacher notes: using the pages (DRAFT)

How the checks were tested

A script tried every single wrong answer and every wrong arrangement on every page. Each check rejected at least 95% of them. A wrong answer that spells any of about 242 000 English words counts as not rejected. Two wrong answers at once were not all tested.

Using the pages

- **Time.** Most pages take 5 to 15 minutes; the colour-by-code page (S1-05) takes longer because pupils colour a picture. Pupils need their own crayons (five colours: yellow, red, blue, green, orange). For a pencil-only copy the pupil writes Y, R, B, G or O in each part; the tally and the matching parts are checked the same way. Light crayons work best: a dark crayon hides the number in the part on a photocopy.
- **If the check fails.** The pupil goes back, looks for the slip, and asks a friend or you if still stuck.
- **Problem solving.** Some pages (S1-07, S1-13, S1-16, S1-20) practise the kinds of question that appear in school tests, such as two-step stories with fractions, with measures and with decimals, and finding a missing number, rather than one numbered syllabus outcome. The key tags them problem solving.
- **Stars.** One star: one skill, as usually taught. Two stars: the same topics with two steps or two clues to combine.
- **Drawings.** Figures of triangles, cuboids, tanks and angles are original drawings. Lengths and angles are not drawn to scale unless a page says so, so a pupil should use the numbers on the figure, not a ruler or a protractor.
- **Not covered.** A printed page cannot check a pupil who draws (a cuboid on an isometric grid, a triangle of a given area) or who builds with real cubes. These verbs of the syllabus are practised only by reading drawings; the plan lists them. Ratio, average, pie charts, fraction division, finding the whole from a percentage, percentage change, circles and algebra are Primary 6 or Primary 4 content in the syllabus summary and are not used.

Photocopying

All pages are black and white with solid lines, so they copy cleanly. Each page is one side only. Pupils write on the page: the boxes are large enough for handwriting. Use A4 and print at 100% (actual size). The small figures in the route and colour pages need a clear copy: use normal toner, not a light setting.

Pictures are not actual size.

Contents: Semester 1

Page	Code	Puzzle	Tier
S1-01	P5-01	Millions	1 star
S1-02	P5-02	Seven Digits	2 stars
S1-03	P5-03	Zero Hunt	1 star
S1-04	P5-04	Multiples Route	2 stars
S1-05	P5-05	Colour the Rocket	1 star
S1-06	P5-06	Brackets Match	2 stars
S1-07	P5-07	Both Sides Equal	2 stars
S1-08	P5-08	Sharing Equally	1 star
S1-09	P5-09	Fractions as Decimals	1 star
S1-10	P5-10	Mixed Number Magic	2 stars
S1-11	P5-11	Mixed Number Route	1 star
S1-12	P5-12	Fraction Chain	1 star
S1-13	P5-13	Fraction of a Quantity	1 star
S1-14	P5-14	Fraction Products	2 stars
S1-15	P5-15	Big Products	2 stars
S1-16	P5-16	Remainder Stories	2 stars
S1-17	P5-17	Point Shift	1 star
S1-18	P5-18	True or False	2 stars
S1-19	P5-19	Measures in Decimals	1 star
S1-20	P5-20	Unit Switch	2 stars

Pages are independent. The footer code shows the page in this booklet and, in brackets, its number in the full set of 40 (P5-01 to P5-40).

What the booklet covers

Whole Numbers: numbers to 10 million reading and writing numbers up to 10 million **S1-01, S1-02**

Whole Numbers: multiply and divide by 10, 100, 1000 and multiples multiplying and dividing by 10, 100, 1000 and their multiples, without a calculator **S1-03, S1-04**

Whole Numbers 2.2 order of operations, without a calculator **S1-05, S1-06, S1-07**

Whole Numbers 2.3 use of brackets, without a calculator **S1-05, S1-06, S1-07**

Fractions 1.1 dividing a whole number by a whole number, with a fraction as the quotient **S1-08**

Fractions 1.2 expressing fractions as decimals **S1-09**

Fractions 2.1 adding and subtracting mixed numbers **S1-10, S1-11**

Fractions 2.2 multiplying a proper or improper fraction by a whole number **S1-12, S1-13, S1-16**

Fractions 2.3 multiplying a proper fraction by a proper or improper fraction **S1-14, S1-15, S1-16**

Fractions 2.4 multiplying two improper fractions **S1-15**

Fractions 2.5 multiplying a mixed number by a whole number **S1-15**

Decimals 1.1 multiplying and dividing decimals (up to 3 decimal places) by 10, 100, 1000 and their multiples **S1-17, S1-18**

Decimals 1.2 converting a measure to a larger or smaller unit in decimal form (km and m, m and cm, kg and g, ℓ and mℓ) **S1-19, S1-20**

Outcomes practised in the other booklet: Percentage: a part of a whole as a percentage; Percentage: a percentage part of a whole; Percentage 1.4; Rate; Area of triangle: base, height and area; Area of triangle: composite figures; Volume 2.1; Volume: cubic units (cm³, m³); Volume 2.3; Volume 2.4; Volume 2.5; Volume 2.6; Geometry 1.1; Geometry 1.2; Geometry 1.3; Geometry 1.4; Geometry: properties of triangles; Geometry: angle sum of a triangle; Geometry: parallelogram, rhombus, trapezium.